NAIVE OBSTRUCTION THEORIES GIVE QUASI-ISOMORPHISMS ON FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4476-4488 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 4490-4504, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 43 of queue_artin.json (pb=631, stmt 4476-4488, proof 4490-4504).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07YP: sources/stacks-artin/pinned-original-source.tex, source lines 4408–4434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YS: sources/stacks-artin/pinned-original-source.tex, source lines 4430–4434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YK: sources/stacks-artin/pinned-original-source.tex, source lines 4150–4153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S and $\mathcal{X}$ be as in Definition 07YP and let $\mathcal{X}$ be endowed with a naive obstruction theory. Let $A \to B$ and $y \to x$ be as in (07YS). Let k be a B-algebra which is a field. Then the functoriality map $E_x \to E_y$ induces bijections*

$$H^i(E_x \otimes^{\mathbf{L}}_A k) \to H^i(E_y \otimes^{\mathbf{L}}_B k)$$

for $i = 0, 1$.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $z = x|_{\mathrm{Spec}(k)}$. Then (RS*) implies that

$$\mathit{Lift}(x, A[k]) = \mathit{Lift}(z, k[k]) \quad \text{and} \quad \mathit{Lift}(y, B[k]) = \mathit{Lift}(z, k[k])$$

because $A[k] = A \times_k k[k]$ and $B[k] = B \times_k k[k]$. Hence the properties of a naive obstruction theory imply that the functoriality map $E_x \to E_y$ induces bijections $\mathrm{Ext}^i_A(E_x, k) \to \mathrm{Ext}^i_B(E_y, k)$ for $i = -1, 0$. By Lemma 07YK our maps $H^i(E_x \otimes^{\mathbf{L}}_A k) \to H^i(E_y \otimes^{\mathbf{L}}_B k)$, $i = 0, 1$ induce isomorphisms on dual vector spaces hence are isomorphisms. $\square$